

Collisions

Conservation of momentum

YOUR AIM TODAY

Model one-dimensional collisions with signed velocities.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

During a short collision, external impulse is often negligible, so total momentum is conserved.

- Total momentum before = total momentum after.
- Choose one positive direction and sign every velocity.
- If objects stick together, they share one final velocity.
- Momentum is conserved; kinetic energy need not be.
- Check whether the sign of the answer matches the described direction.

MEMORY RULE

Write one before = after momentum equation with signs visible.

Stick together

WORKED STEPS

2 kg at 5 m/s hits 3 kg at rest and sticks: $2(5) = (2+3)v$, so $v = 2$ m/s.

Opposite directions

WORKED STEPS

Right is positive, so a particle moving left has negative velocity.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. 4 kg at 6 m/s hits 2 kg at rest and sticks. Find v.	
2. Find KE lost in Q1.	
3. 3 kg at +4 m/s collides with 1 kg at -2 m/s; they stick. Find v.	
4. 2 kg at 5 m/s hits 3 kg at rest. Afterward 2 kg moves at 1 m/s. Find other velocity.	
5. A stationary object explodes into 2 kg at +6 m/s and 3 kg. Find the 3 kg velocity.	
6. Why is kinetic energy not automatically conserved?	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$24=6v$, so $v=4$ m/s.
2	Before 72 J; after 48 J; loss 24 J.
3	$12-2=4v$, so $v=2.5$ m/s.
4	$10=2+3v$, so $v=8/3$ m/s.
5	$0=12+3v$, so $v=-4$ m/s.
6	Some energy may become heat, sound or deformation.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.